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Deep-ocean heat uptake and equilibrium climate response

Chao Li · Jin-Song von Storch · Jochem Marotzke

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Abstract We integrate the coupled climate model ECHAM5/MPIOM to equilibrium under atmospheric CO_2 quadrupling. The equilibrium global-mean surface-temperature change is 10.8 K. The surface equilibrates within about 1,200 years, the deep ocean within 5,000 years. The impact of the deep ocean on the equilibrium surface-temperature response is illustrated by the difference between ECHAM5/MPIOM and ECHAM5 coupled with slab ocean model (ECHAM5/SOM). The equilibrium global-mean surface temperature response is 11.1 K in ECHAM5/SOM and is thus 0.3 K higher than in ECHAM5/MPIOM. ECHAM5/MPIOM shows less warming over the northern-hemisphere mid and high latitudes, but larger warming over the tropical ocean and especially over the southern-hemisphere high latitudes. ECHAM5/MPIOM shows similar polar amplification in both the Arctic and the Antarctic, in contrast to ECHAM5/SOM, which shows stronger polar amplification in the northern hemisphere. The southern polar warming in ECHAM5/MPIOM is greatly delayed by Antarctic deep-ocean warming due to convective and isopycnal mixing. The equilibrium ocean temperature warming under CO_2 quadrupling is around 8.0 K and is near-uniform with depth. The global-mean steric sea-level rise is 5.8 m in equilibrium; of this, 2.3 m are due to the deep-ocean warming after the surface temperature has almost equilibrated. This result suggests that the surface temperature change is a poor predictor for steric sea-level change in the long term. The effective climate response method described in Gregory et al. (2004) is evaluated with our simulation, which shows that their method to estimate the equilibrium climate response is accurate to within 10 %.

Keywords Deep-ocean heat uptake · Equilibrium climate response · Climate sensitivity

1 Introduction

In response to changes in external forcing, the climate system adjustment processes involve timescales ranging from days in the atmosphere to several millennia in the deep ocean. The long-term adjustment with timescales above a millennium is not well understood, because rarely have atmosphere-ocean global circulation models (AOGCM) been run to equilibrium. To gain a more comprehensive understanding of the long-term adjustment of the deep ocean and its impact on the surface equilibrium response, we integrate a coupled climate model, ECHAM5/MPIOM, to equilibrium under atmospheric CO_2 quadrupling. The final equilibrium in ECHAM5/MPIOM is then compared with a simulation of ECHAM5 coupled to a slab ocean model (ECHAM5/SOM).

Equilibrium climate sensitivity (ECS), as defined by the Intergovernmental Panel of Climate Change (IPCC) assessments (e.g. Cubasch et al. 2001; Randall et al. 2007), is the equilibrium annual and global mean surface temperature response to atmospheric CO_2 doubling from a pre-industrial level. In this study, to avoid confusion, we define the equilibrium climate response (ECR) as the equilibrium annual and global mean surface temperature response to atmospheric CO_2 quadrupling. To reduce the computational cost of an AOGCM integrated to equilibrium,

C. Li (✉) · J.-S. von Storch · J. Marotzke
Max Planck Institute for Meteorology, Bundesstrasse 53, 20146 Hamburg, Germany
e-mail: chao.li@zmaw.de

C. Li
International Max Planck Research School on Earth System Modeling, Hamburg, Germany

different methods have been developed to estimate the ECR or ECS. In one of the methods, the ECR is obtained using an AGCM coupled with a slab ocean model (SOM) and the thermodynamic part of the sea-ice component. The SOM is a simple non-dynamic model of the upper ocean with prescribed ocean heat transport convergence. This simplified configuration makes it possible to estimate the ECR with only several decades of integration. But the disadvantage is that any change in the ocean heat transport cannot be represented. Another method estimates the ECR from the transient climate response in an AOGCM. Gregory et al. (2004) show that, when the net downward TOA (top of the atmosphere) radiative heat flux N is plotted against the surface temperature change ΔT with a fixed forcing of the Hadley Center slab climate model version 3 (HadSM3), a straight line gives a good fit. Hence, they suggest that a linear regression between N and ΔT gives a good estimate of the ECR and the radiative forcing. However, the feedback strength is time-dependent, which is associated with difference in cloud feedback arising from inter-hemispheric temperature differences due to the slower warming rate of the Southern Ocean (Senior and Mitchell 2000), and in a more complex climate model, some feedbacks will change with the climate state (Boer and Yu 2003; Gregory and Webb 2008). Therefore, the assumptions of a linear feedback is valid only for perturbations of a few degrees, and the method of Gregory et al. (2004) should be evaluated with an AOGCM.

Williams et al. (2008) and Winton et al. (2010) show that the spatial warming pattern changes as the climate evolves toward equilibrium because of the ocean heat uptake. Apart from the impacts on the spatial pattern of climate change, the ocean heat uptake also influences the temporal aspects of climate change from decades to multi-millennia. The large upper-ocean heat uptake changes the climate transient response on decadal timescales, and the small deep-ocean heat uptake (DOHU) may influence the ECR on timescales from centuries to millennia due to its impact on some slow feedbacks of the climate system (Knutti and Hegerl 2008). However, the long-term effects of DOHU are poorly understood, because it requires thousands of years of simulation to achieve a steady state of an AOGCM.

Beside the warming processes on the earth surface, the ocean warming is also very important in sea level change and ocean biogeochemical processes. Levitus et al. (2005) showed that roughly 80 % of the earth radiation imbalance due to the anthropogenic greenhouse gases has gone into heating the ocean. In response to this external forcing, the heat content of the world ocean above 3,000 m depth has increased by nearly 2×10^{23} joules, which corresponded to a warming rate of 0.3 W m^{-2} in the last half of the 20th century (Levitus et al. 2000). By using different historical data, the ocean heat uptake of the top 700 m ocean is estimated as 0.41 W m^{-2} (Levitus et al. 2009) and 0.64 W m^{-2} (Lyman et al. 2010); von Schuckmann and Traon (2011) found 0.55 W m^{-2} for the top 1,500 m of the ocean. The ocean heat uptake below 2,000 m is estimated as 0.07 W m^{-2}. At present, the upper-ocean warming is larger than the deep-ocean warming, because the upper ocean is ventilated much more efficiently. However, will the deep ocean also experience less warming than the upper ocean in the final equilibrium? And how does the deep-ocean warming contribute to the global sea level change in the long term? A long integration with an AOGCM is required to answer these questions.

Very rarely have coupled AOGCMs been integrated over more than 1,000 years with atmospheric CO_2 doubling or quadrupling. The first long run with a coupled AOGCM was described by Stouffer and Manabe (1999) and Stouffer and Manabe (2003). By using a flux-adjusted coupled AOGCM with coarse resolution, they performed a 15,000-year pre-industrial control run and a 4,000-year integration with atmospheric CO_2 doubling. In their experiments, the Atlantic meridional overturning circulation (AMOC) recovered to the pre-industrial control level, implying that the ocean heat transport showed little change. By using an AOGCM with flux-adjustment, ECHAM3/LSG, Voss and Mikolajewicz (2001) performed two 850-year integrations with atmospheric CO_2 doubling and quadrupling. They mainly focused on describing the long-term climate adjustment processes and their timescales. Gregory et al. (2004) analyzed a 1,200-year integration with atmospheric CO_2 quadrupling by using the non-flux-adjusted Hadley Center Coupled Model, version 3 (HadCM3), but the ocean was still warming up at the end of the run. Recently, by using the low-resolution version of the non-flux-adjusted Community Climate System Model, version 3 (CCSM3), Danabasoglu and Gent (2009) analyzed a 3,000-year long integration with atmospheric CO_2 doubling, but the total ocean heat content still had a small trend. They concluded that the equilibrium climate sensitivity to CO_2 doubling determined by a SOM is similar to that in an AOGCM, because the AMOC and northward ocean heat transport did not change much in response to CO_2 doubling in CCSM3. In our long integration, however, the AMOC is weaker by 46 % in the final equilibrium, which gives us an opportunity to study the impact of a deep-ocean circulation change on the surface equilibrium.

In order to gain a more comprehensive understanding of the deep ocean's role on climate response to atmospheric CO_2 forcing, we integrate ECHAM5/MPIOM to equilibrium under atmospheric CO_2 quadrupling. We study the slow deep-ocean warming, the change in deep-ocean circulation, and their impact on the surface equilibrium response. Section 2 is a brief introduction of the models and experiment design. Section 3 shows the long-term

adjustment toward equilibrium. Section 4 compares the final equilibrium in the ECHAM5/MPIOM with that in ECHAM5/SOM; Sect. 5 discusses the effective climate response; and Sect. 6 gives our conclusions.

2 Models and experiment design

The coupled climate model applied in this study is a coarse-resolution version of ECHAM5/MPIOM. The spectral atmospheric model ECHAM5 is run at T31 resolution ($\sim$3.75°) with 19 levels (Roeckner et al. 2003, 2006). The Max-Planck-Institute Ocean Model (MPIOM) is used at a resolution of roughly 3° near the equator with 40 levels. Technical details of MPIOM and the embedded sea ice model can be found in (Marsland et al. 2003) and (Jungclaus et al. 2006). Here, we summarize the main features. MPIOM is a z-coordinate global GCM based on the primitive equations for a hydrostatic Boussinesq fluid with a free surface. The spatial arrangements of scalar and vector variables are formulated on an orthogonal curvilinear C-grid (Arakawa and Lamb 1977). The coordinate poles are located on central Greenland and Antarctica, thus avoiding the pole-singularity problem. The model uses an along-isopycnal diffusion following Redi (1982) and Griffies (1998), and isopycnal tracer transport by unresolved eddies is parameterized following Gent et al. (1995). For the vertical eddy viscosity and diffusion the Richardson-number dependent scheme of Pacanowski and Philander (1981) is applied. Since the Pacanowski–Philander (PP) scheme in its classical form underestimates the turbulent mixing close to the surface, an additional wind mixing parameterization is included. In the presence of static instability, convective overturning is parameterized by greatly enhanced vertical diffusion. A bottom boundary layer slope convection scheme allows for an improved representation of the flow of statically unstable dense water over sills. The embedded sea ice module is a Hibler-type dynamic-thermodynamic sea ice model with viscous-plastic rheology and snow (Hibler 1979). Thermodynamic growth of sea ice is described by the zero-layer formulation of Semtner (1976). Ocean and atmosphere are coupled daily using the OASIS3 coupler (Valcke et al. 2003); no flux adjustments are applied. A higher-resolution version of the model (Jungclaus et al. 2006) has been used for the scenario simulations of the IPCC assessment report 4 (AR4).

To make our model run at hot climate states, we employ two modifications of ECHAM5. The first is an adjustment of the prescribed and fixed ozone concentration based on the observed climatology in standard ECHAM5. To avoid the artificially large ozone concentration in the upper troposphere that would follow an increase in tropopause height, the ozone contributing to a concentration above 0.15 ppmv below the tropopause is moved into the uppermost level in ECHAM5. Second, a positive definite optical thickness in the longwave radiative transport model is ensured, to avoid unphysical negative optical thicknesses for gases and aerosols. Both modifications were introduced for the simulation of the Paleocene–Eocene Thermal Maximum by Heinemann et al. (2012). Without the ozone adaptation, we find an unstoppable global warming after 5,000 years of integration in ECHAM5/MPIOM, and the global surface temperature is 4.0 K higher than the experiment which is used in this study. Both modifications hardly affect the pre-industrial control simulation.

The SOM assumes a fixed mixed-layer depth (default value is 50 m). The SST is determined by the energy balance between net surface heat fluxes and a Q-flux that approximates the ocean heat transport convergence. In the SOM, the Q-flux values are diagnosed from an ECHAM5 control run and represent seasonal deep water exchange and horizontal ocean heat transport. Hence, the ocean heat transport in ECHAM5/SOM should not change in response to atmospheric CO_2. However, the Q-flux term is only applied in the ice-free regions; in the ice-covered region the surface heat flux is calculated differently by the simple sea-ice model. This can lead to an unrealistic change of the surface heat flux during the loss of Antarctic sea-ice, and thus change the implied ocean heat transport in the Southern Hemisphere.

Our first integration with ECHAM5/MPIOM is a 1,600-year pre-industrial control run (CNTR) with constant atmospheric CO_2 at 278 ppmv. CNTR is close to a steady-state climate at the end of the simulation. The last 100 years of data are used as the CNTR reference in this study. A second integration starts from the end of CNTR; the atmospheric CO_2 concentration is increased by 1 % per year until it reaches CO_2 quadrupling at year 140, and is held constant thereafter. The experiment is continued for a further 5,940 years until the whole system has reached the final equilibrium. To evaluate the signal of climate change at any time in these integrations, the CNTR is subtracted. In our third integration, ECHAM5 is coupled to a 50 m slab ocean model (SOM). We first run a 110-year atmosphere-only spin-up under preindustrial boundary conditions and 278 ppmv pCO2; the ocean advective heat flux is diagnosed from the last 50 years of this spin-up. Then we run the model for another 50 years as the pre-industrial control run of ECHAM5/SOM. Thereafter, we increase pCO2 by 1 % per year until it reaches 2 $\times$ CO_2 at year 70 and 4 $\times$ CO_2 at year 140, and it held constant thereafter. The experiment is continued for a further 660 years; we use averages over the last 100 years for our analysis.

In this study, we performed idealized experiments with prescribed atmospheric CO_2 concentrations that are kept constant over long periods of time. The long-term change of the atmospheric CO_2 concentration due the carbon cycle

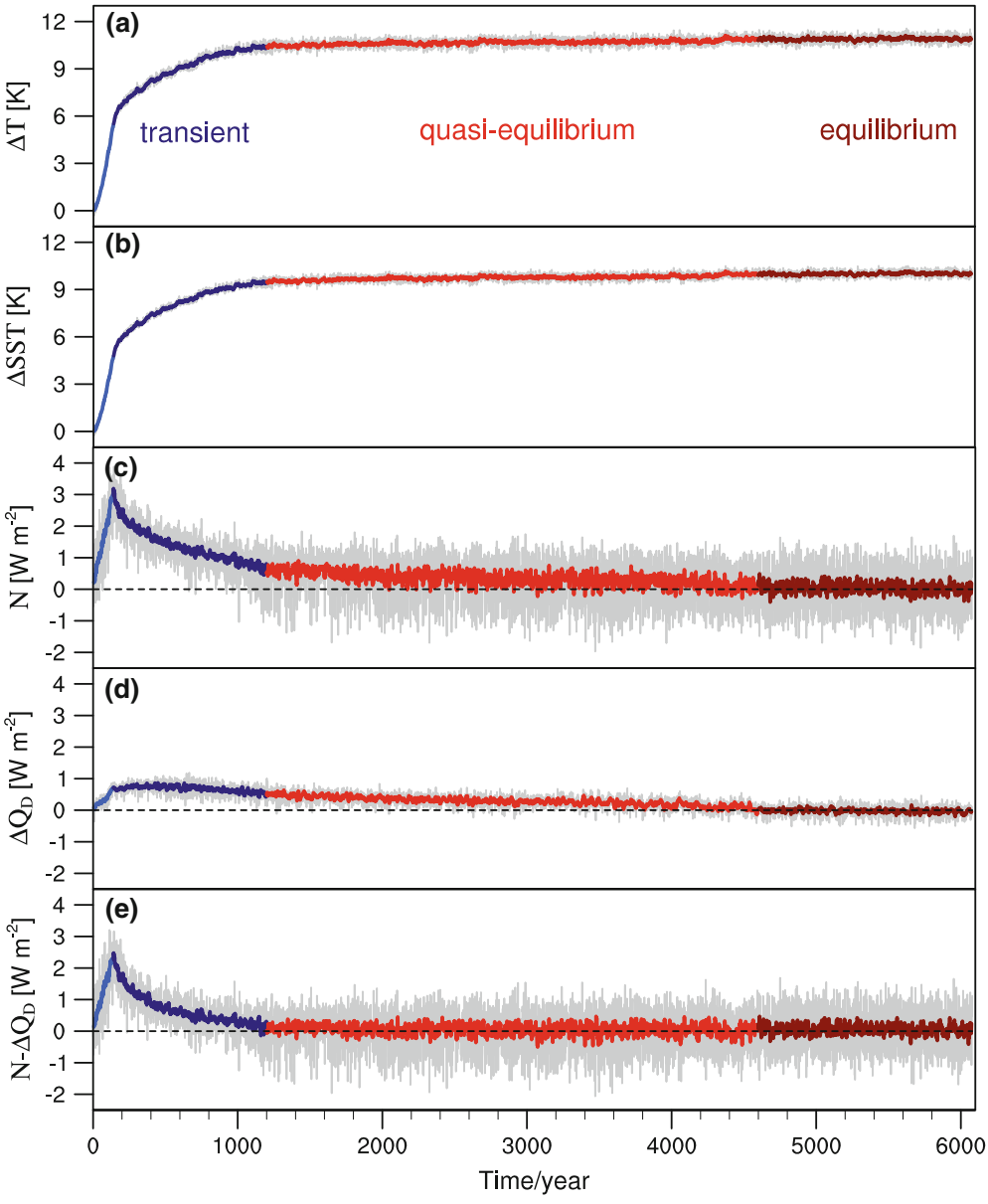


Fig. 1 Time series of, **a** globally averaged surface temperature change (ΔT, Unit: K), **b** sea surface temperature change (ΔSST, Unit: K), **c** globally averaged net TOA downward radiative flux (N, Unit: W m^{-2}), **d** globally averaged deep-ocean heat uptake below 1,525 m (ΔQ_D, Unit: W m^{-2}), **e** difference between **c** and **d** (N-ΔQ_D, Unit: W m^{-2}), by using ECHAM5/MPIOM with CO_2 quadrupling in 140 year. All time series are relative to the reference period from CNTR. The *gray line* represents annual-mean data for the entire experiment, the *other colors* represent an 11-year running mean

processes in the earth system has not been included. Moreover, the coupled climate model used here has neither dynamic glaciers nor dynamic vegetation; as one consequence, we cannot explore the impact of ice-sheet melting on the AMOC and other climate components. Our idealized experimental strategy allows us, however, to explore the true equilibrium response of a climate model to enhanced CO_2 concentration.

3 Multi-millennium adjustment toward equilibrium

3.1 Adjustment stages

The equilibrium global-mean surface temperature response to atmospheric CO_2 quadrupling is 10.8 K (Fig. 1a). After year 140, the time of CO_2 quadrupling, the globally averaged surface air temperature has increased by approximately 5.5 K. In the following millennium with atmospheric CO_2 constant at 1,112 ppmv, the globally averaged surface temperature increases by an additional 5.3 K. There is still a very weak trend after the first 1,200 years integration; the surface air temperature increases by 0.4 K in 3,400 years. This weak trend mainly comes from the Southern Ocean and the Antarctic continent (Fig. 2d) where the surface warming is closely connected to the deep-ocean heat uptake in the Southern Ocean (Gregory 2000). In the ocean, the globally averaged equilibrium surface temperature response to CO_2 quadrupling is 10.0 K (Fig. 1b), 92 % of the globally averaged surface temperature change.

The net TOA downward radiative flux (N) reaches its maximum of 3.5 Wm^{-2} at year 140 when the atmospheric

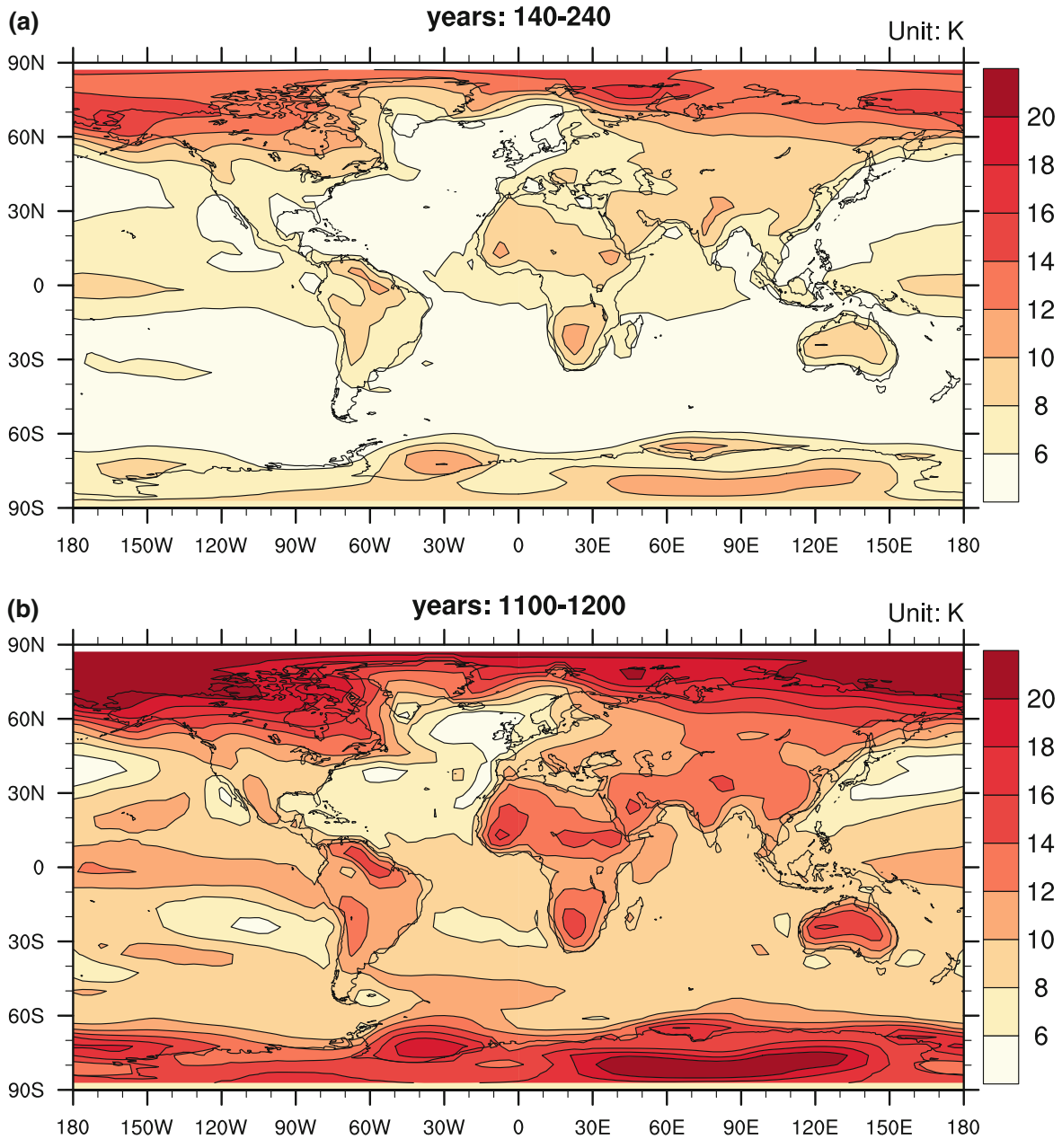


Fig. 2 Geographic distribution of the surface air temperature change of ECHAM5/MPIOM in response to CO_2 quadrupling, **a** by the beginning of the transient period (i.e., the average from the 140th year to the 240th year), **b** the end of the transient period (i.e., the average from the 1,100th year to the 1,200th year), **c** the end of the equilibrium period (i.e., the average from the 5980th year to the 6,080th year). **d** is the difference between **c** and **b**. Unit: K

CO_2 concentration has just quadrupled (Fig. 1c). In the following millennium, N decreases but is still 0.7 W m^{-2} at year 1,200 when the surface temperature change almost becomes stationary; N goes to zero when the system reaches the final equilibrium. The DOHU is non-zero at year 1,200, but it goes to zero when the deep-ocean reaches equilibrium (Fig. 1d). Hence, the surface temperature change reaches equilibrium when N is balanced by the ocean heat transfer across 1,500 m, rather than when the net TOA radiative flux goes to zero.

According to the adjustments of the global-mean surface temperature, of the radiative flux into the whole system, and of the heat flux into the deep ocean (Fig. 1), the time after CO_2 quadrupling can be divided into three periods: the transient period (140–1,200 year), the quasi-equilibrium period (1,200–4,600 year) and the equilibrium period (4,600–6,080 year). In the transient period, the surface temperature and the upper ocean temperature are non-stationary, the whole system is adjusting. In the quasi-equilibrium period, the surface temperature and the upper

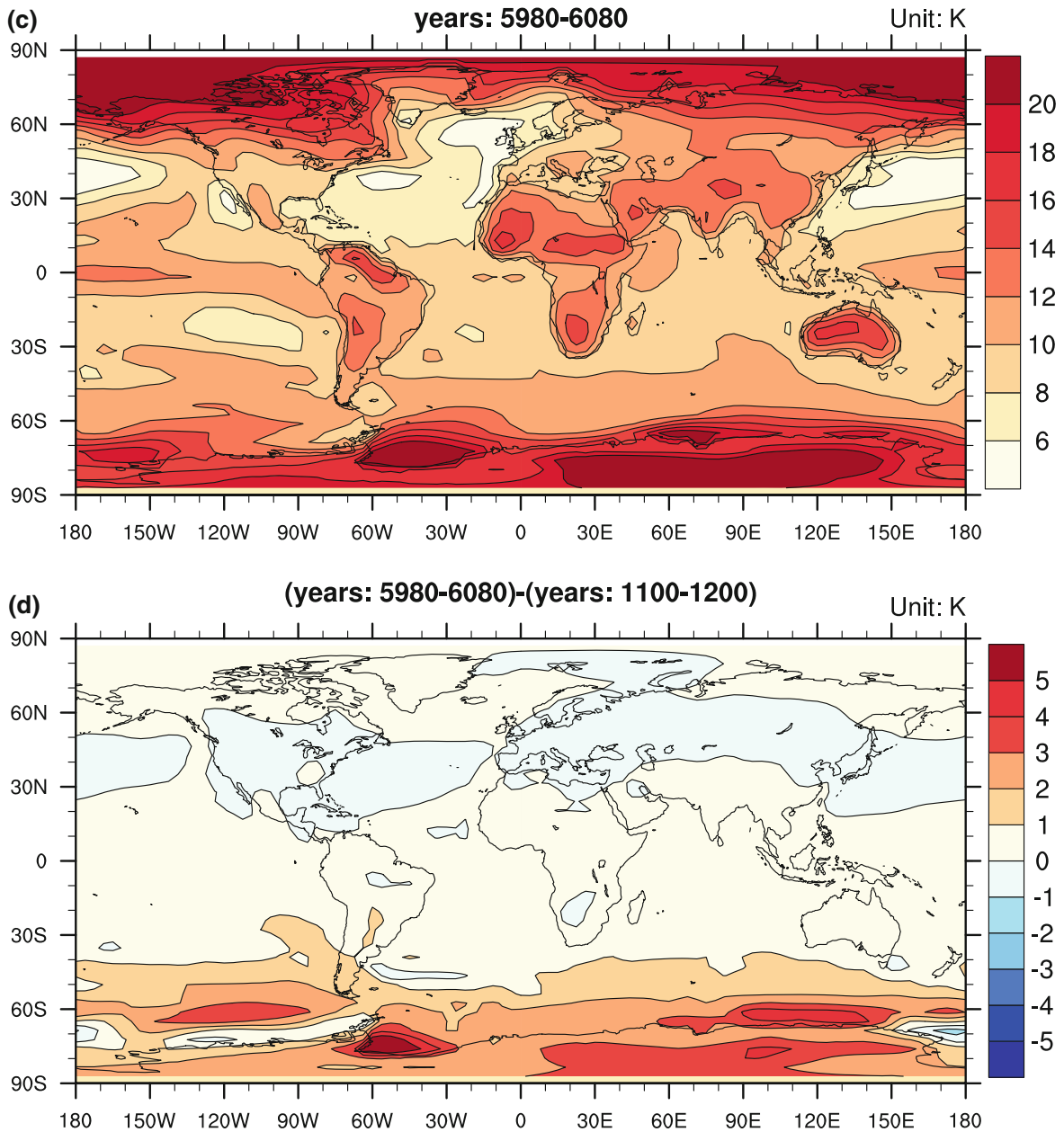


Fig. 2 continued

ocean temperature (above 1,500 m) are nearly stationary, but the deep ocean is still warming. In the equilibrium period, the surface, the upper ocean, and the deep-ocean temperature change are stationary, and the whole system has reached its final equilibrium under atmospheric CO_2 quadrupling.

3.2 Patterns of surface temperature change

We now compare the patterns of surface temperature change that have occurred by the beginning of the transient period, the end of the transient period, and the end of the equilibrium period (Fig. 2a). By the beginning of the transient period, we see the familiar pattern of greater warming over land than over the ocean, large warming in the Arctic, and less warming over the North Atlantic and the Southern Ocean (e.g. Meehl et al. 2007). By the end of the transient period, the surface temperature has almost reached equilibrium, except over the Southern Ocean and Antarctica (Fig. 2a, b, c) where the surface temperature still rises during the quasi-equilibrium period. The slow surface warming over these domains is related to the DOHU (Gregory 2000). The greater warming in southern high latitudes during the quasi-equilibrium period (Fig. 2a, c) implies that in equilibrium, our AOGCM shows similar polar amplification in both the Arctic and Antarctic

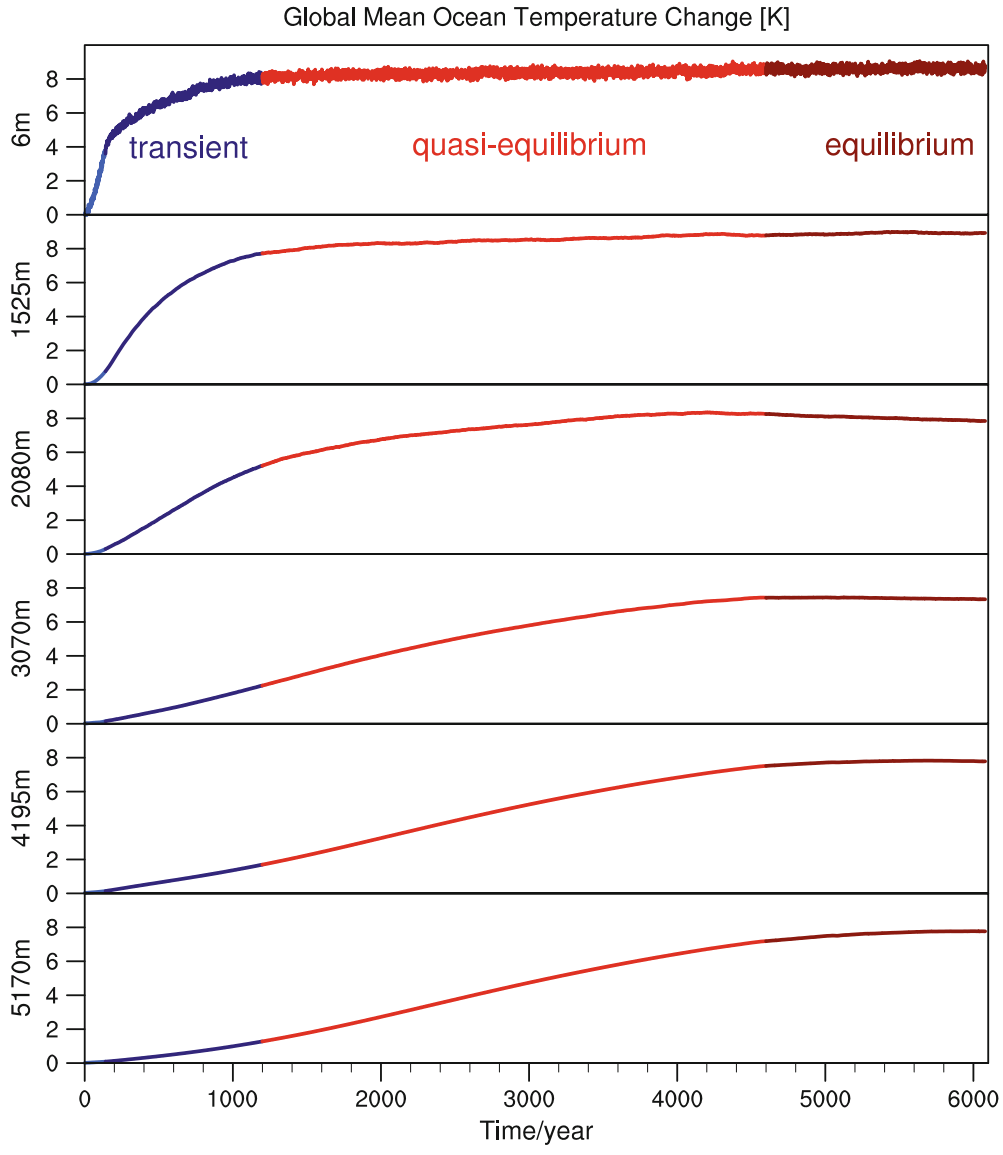


Fig. 3 Time series of annual-mean, globally averaged ocean temperature change with CO_2 quadrupling in ECHAM5/MPIOM. The time series are relative to the reference period from CNTR. Unit: K

domains. The southern polar warming is greatly delayed by the Southern Ocean deep-ocean warming, implying that shorter simulations only show polar amplification in the Northern Hemisphere (e.g. Manabe and Stouffer 1980; Holland and Bitz 2003).

3.3 Ocean temperature and sea level change

The upper ocean (above 1,500 m) and the deep ocean (below 1,500 m) adjust to equilibrium on two different time scales (Fig. 3). The upper ocean reaches equilibrium by the end of the transient period, just as the surface (Fig. 1a). During the CO_2 increase period, the response of the deep ocean is very weak, and the warming trend increases in the following centuries. The deep ocean is still far away from the new equilibrium when the upper ocean is already stable, implying that the deep ocean still takes up heat from the upper ocean when the surface temperature change is almost stationary. The whole system reaches the final equilibrium after about 4,600 years when the deep ocean stops warming.

At equilibrium, we find a near-uniform warming of around 8.0 K at almost all levels of the ocean (Fig. 4). The maximum warming, which is about 10.0 K, is located between 1,000 and 2,000 m depth. In order to estimate the ocean equilibrium response to atmospheric CO_2 forcing, researchers have previously mainly used upwelling-diffusion energy-balance climate models (Hoffert et al. 1980; Gornitz et al. 1982; Harvey and Schneider 1985; Wigley and Raper 1987, 1992; Raper et al. 2001; Marčelja 2010). These models have suggested small warming in the deep ocean but large warming in the surface. Therefore, the ocean heat content change is mainly determined by the temperature change occurring in the upper 1,000 m

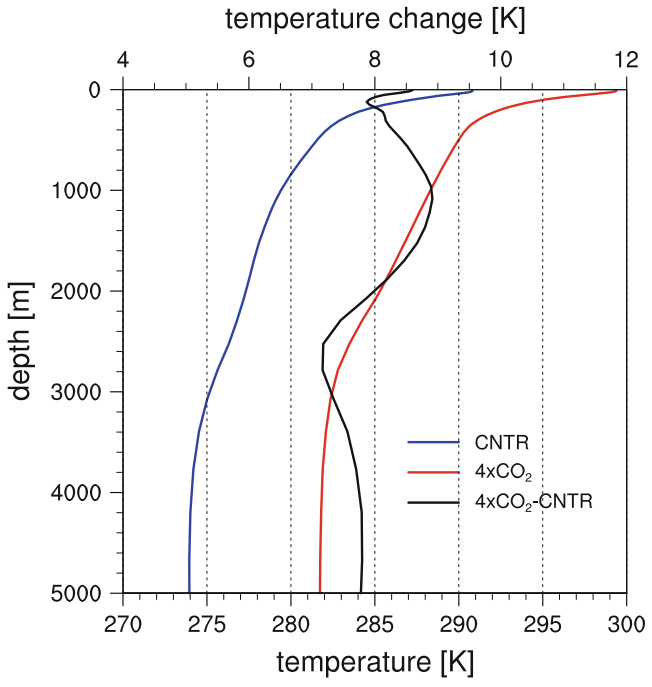


Fig. 4 Globally averaged ocean temperature profiles. The *blue line* represents the pre-industrial control run, the *red line* represents the final 100-year mean of the 4 × CO_2 run, and the *black line* represents the difference between the final 100 years of 4 × CO_2 relative to CNTR. Unit: K

(Marčelja 2010). In contrast, Harvey and Schneider (1985) used a three-layer ocean box model coupled to a global-mean energy balance model and found uniform ocean warming under an increase of the solar constant. Our AOGCM results confirm the results of the globally averaged ocean box model but are inconsistent with the results of the upwelling-diffusion energy-balance ocean models. It is conceivable that the results of upwelling-diffusion models can be brought into agreement with our results by including a time-dependent bottom temperature. This strategy was pursued by Harvey and Schneider (1985) (Sect. 5), who, however, did not nearly increase their bottom temperature enough to be consistent with our results.

Since the deep ocean and the upper ocean show a nearly identical temperature change in the final equilibrium, the deep-ocean temperature change also plays an important role for the long-term globally averaged sea level change. Notice that our model cannot represent a possible sea level rise due to an additional fresh-water input caused by melting land ice. But following Landerer et al. (2007), we calculate the steric sea level change (ζ_S), thermosteric (ζ_S^{thermo}) and halosteric (ζ_S^{halo}) sea level changes contributed by each model depth layer from

$$\zeta_S(K) = \frac{1}{A}\int_A \frac{\rho(S_C,T_C,p) - \rho(S_{EX},T_{EX},p)}{\rho(S_C,T_C,p)}\Delta Z dxdy, \quad (1)$$

$$\zeta_S^{\text{thermo}}(K) = \frac{1}{A}\int_A \frac{\rho(S_C,T_C,p) - \rho(S_C,T_{EX},p)}{\rho(S_C,T_C,p)}\Delta Z dxdy, \quad (2)$$

$$\zeta_S^{\text{halo}}(K) = \frac{1}{A}\int_A \frac{\rho(S_C,T_C,p) - \rho(S_{EX},T_C,p)}{\rho(S_C,T_C,p)}\Delta Z dxdy, \quad (3)$$

where the subscripts *EX* and *C* refer to the atmospheric CO_2 quadrupling experiment and to the control run, ΔZ is the thickness of each model layer, A is the ocean area at each model level, and the in-situ density ρ is a time-dependent, nonlinear function of salinity, temperature and pressure (Gill 1982). ζ_S, ζ_S^{thermo} and ζ_S^{halo} are calculated as a function of model level K. The thermosteric (ζ_S^{thermo}) and halosteric (ζ_S^{halo}) sea level changes are calculated to evaluate the individual contribution of temperature and salinity to the steric anomalies in reference to the temperature and salinity fields of the control run. The total globally averaged sea level change is cumulatively summed up over the whole ocean depth, the contribution from the upper ocean is cumulatively summed up over the top 1,500 m, and the contribution from the deep ocean is cumulatively summed up over the depth below 1,500 m (Fig. 5a). Global-mean sea level rises by 5.8 m with the 10.8 K surface warming in the steady state of the 4 × CO_2 run, roughly 0.5 m per 1.0 K increase of the surface temperature. This number is much larger than in the upwelling-diffusion ocean model of Marčelja (2010), who only found a sea level rise of 0.125 m for a 1.0 K surface temperature rise. During our transient period, the global sea level change is mainly caused by the upper-ocean warming. However, during the quasi-equilibrium period the sea level change due to the upper-ocean warming is stationary while the sea level change due to deep-ocean warming is still ongoing (Fig. 5). In the final equilibrium, the global sea level change induced by the deep-ocean warming is 3.4 m while the upper ocean contributes 2.4 m.

Starting at the sea surface, the thermosteric and halosteric contribution from each level are cumulatively summed up to a depth of 5,000 m for the last 100 years of the final equilibrium period (Fig. 6). Generally, the thermosteric sea level change plays a dominant role for the global-mean sea level change while the halosteric change is less important; the salinity anomalies may, however, have a significant influence on regional sea level change (e.g. Landerer et al. 2007). Figure 6 also shows that the thermosteric signal has its largest contribution within the upper 4,000 m, not only in the upper 1,000 m.

Future sea-level rise has also been estimated using empirical approaches (Rahmstorf 2007; Vermeer and Rahmstorf 2009). The general assumption of the empirical approach is that the relationship between sea level change

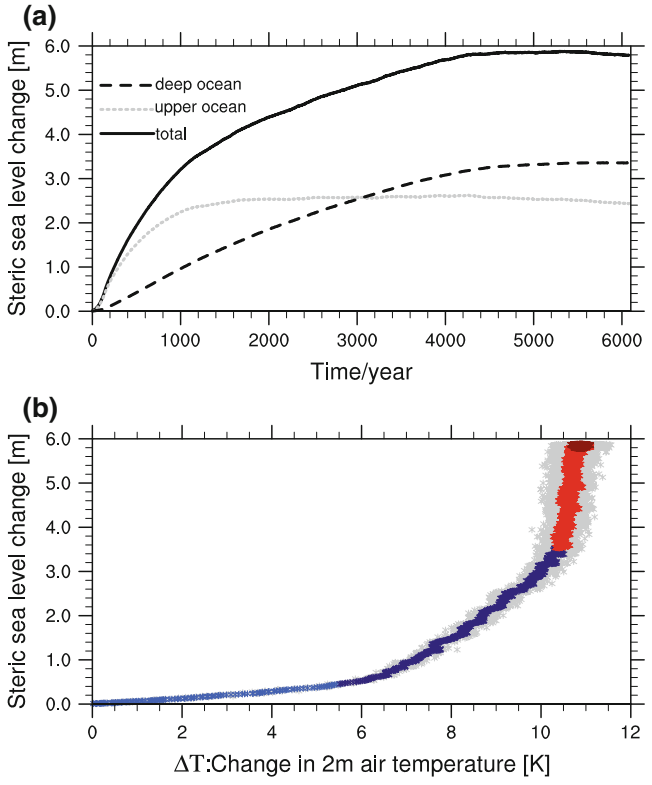


Fig. 5 **a** Time series of the annual-mean, globally averaged steric sea-level change (*black solid line*), and its contribution from the upper ocean (*gray dotted line*) and the deep ocean (*black dashed line*). The separation between upper and deep ocean is at 1,500 m. Unit: m. **b** Globally averaged steric sea-level change (Unit: m) as a function of globally averaged surface temperature change (ΔT) (Unit: K). The *gray crosses* represent annual-mean data for the entire experiment, the *other color crosses* represent data from an 11-year running mean. The *light blue crosses* represent the CO_2 increase period (1–140 year), the *dark blue crosses* represent the transient period (140–1,200 year), the red crosses represent the quasi-equilibrium period (1,200–4,600 year), and the firebrick crosses represent the final equilibrium period (4,600–6,080 year). The change is with respect to the reference period from CNTR

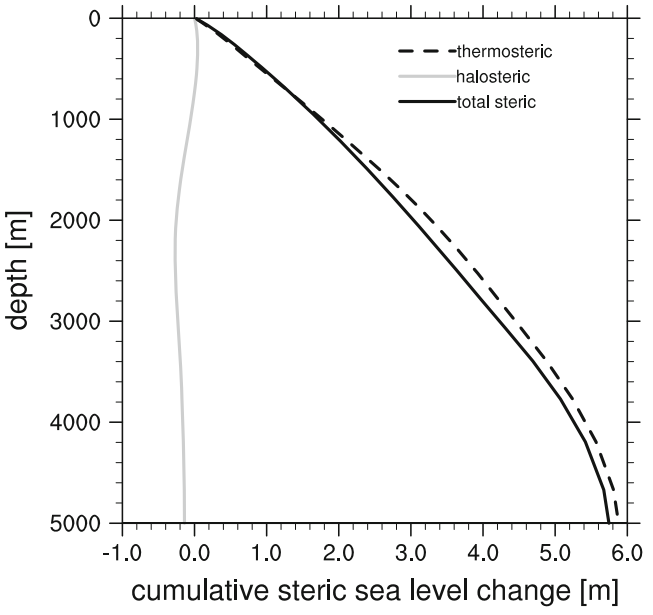


Fig. 6 Cumulative sum of thermosteric sea-level change (*dashed line*), halosteric sea-level change (*grey line*) and total steric sea-level change (*black line*) for the global ocean in the last 100 years of the equilibrium period. Starting at the surface, the steric anomaly from each depth layer is added up. Unit: m

and surface temperature change or radiative forcing will hold in the future and for a much greater range of warming than occurred during the period from which it was calibrated. Based on this assumption, the future sea level change can be calculated directly from the climate model projections of global warming. Our AOGCM result shows that the globally averaged sea level changes by 0.45 m and the surface temperature increases by 5.5 K during the first 140 years when the atmospheric CO_2 increases by 1 % year^{-1}, and the globally averaged sea level rises by 3.05 m and surface temperature increases by 4.9 K during the transient period. During the quasi-equilibrium period, the globally averaged sea level still rises by 2.3 m while the surface temperature is almost stationary (Fig. 5b). The relationship between sea level rise and the surface temperature will not hold because the ocean temperatures at different depths respond on different timescales. During the short-term transient response, we may use the empirical approaches and 1-D upwelling-diffusion ocean models to estimate steric sea level change, but for the long-term equilibrium response, we cannot use these two methods because they both underestimate the contribution of the deep-ocean warming to the steric sea level change.

3.4 Change in Atlantic meridional overturning circulation

As in most previous global warming studies, the Atlantic meridional overturning circulation (AMOC) weakens, but does not collapse in our experiment (Fig. 7). We define an Atlantic meridional overturning circulation index (MOI) as the annual-mean zonally integrated streamfunction in Sverdrups (1 Sv $\equiv 10^6$ m^3 s^{-1}) at the depth of 1,000 m in the Atlantic at 30°N. The MOI weakens from 18.5 Sv to about 7.5 Sv (decline by 62.2 %) during the first 140 years while CO_2 is increasing (Fig. 7c). Thereafter, the MOI slowly recovers and reaches 10 Sv in the new equilibrium. The increase of the AMOC with stationary $4 \times CO_2$ forcing is due to the salt advection feedback as found by Latif et al. (2000). In a warmer climate, the enhanced evaporation in the tropical Atlantic leads to increased salinity there; this saline water is eventually advected into the deep-water formation regions in the North Atlantic, decreases the upper ocean stability there, and thus strengthens the AMOC. But in contrast to other AOGCM long-term integrations (Voss and Mikolajewicz 2001;

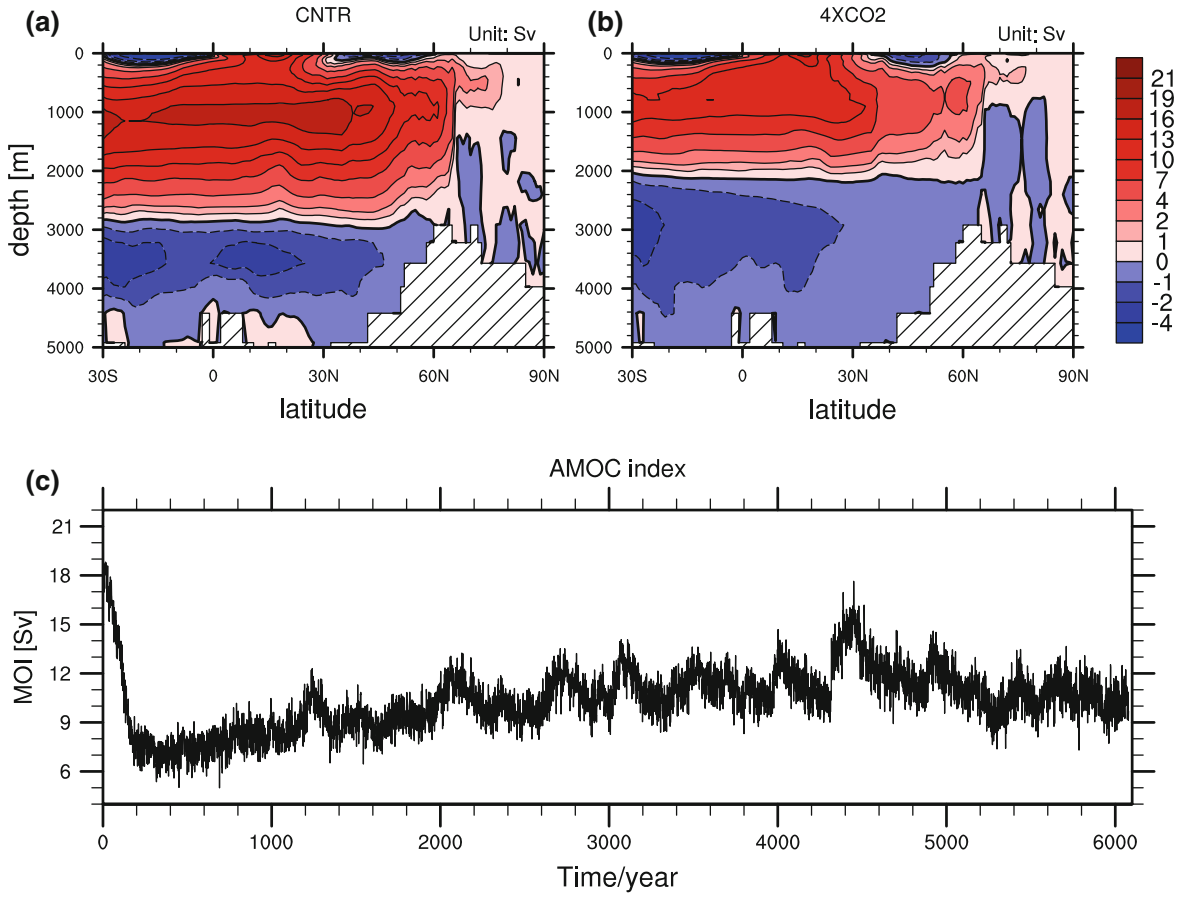


Fig. 7 Latitude-depth distribution of the meridional streamfunction in the Atlantic, **a** for the pre-industrial control run, and **b** for the equilibrium of CO_2 quadrupling. **c** Time series of the Atlantic meridional circulation index (MOI). The MOI is defined as the annual mean zonally integrated streamfunction at the depth of 1,000 m in the Atlantic at 30°N. Unit: Sv

Stouffer and Manabe 2003), the AMOC in our experiment does not recover to the pre-industrial control value. The MOI is about 46.0 % weaker than in CNTR, and the AMOC upper cell is shallower, which indicates that the North Atlantic Deep Water does not penetrate as deep as in CNTR, and the bottom cell becomes thicker and weaker.

4 Equilibrium climate response in ECHAM5/SOM

In ECHAM5/MPIOM the AMOC declines by 46 % in the 4 × CO_2 case; this weakening in AMOC reduces the northward ocean heat transport by 0.57 PW across 30°N (decline by about 35 % relative to the pre-industrial mean value, Fig. 8a). By contrast, the implied ocean heat transport with CO_2 quadrupling is almost the same as in CNTR in ECHAM5/SOM, except in the southern-hemisphere high latitudes where we find an unrealistic decline by 0.29 PW (Fig. 8b). The implied ocean heat transport should be constant by construction but is not really because of a contribution from changes in sea-ice. To investigate whether the surface equilibrium response is affected by the change in ocean heat transport, we compare the ECR in both ECHAM5/MPIOM and ECHAM5/SOM.

In ECHAM5/SOM, the globally averaged surface temperature is 287.6 K in the pre-industrial control run and is 0.9 K higher than that in ECHAM5/MPIOM; SST is also 0.9 K higher. The ECR is 11.1 K in ECHAM5/SOM and is 0.3 K higher than in ECHAM5/MPIOM. The change in SST is identical in both models, which suggests that the slight overestimation in ECHAM5/SOM is mainly caused by the response over the continent (Table 1). The equilibrium climate sensitivity of ECHAM5/SOM with atmospheric CO_2 doubling from 278 to 556 ppmv amounts to 3.7 K, which is in the climate sensitivity range from 2.1 to 4.4 K reported in the IPCC AR4 (Randall et al. 2007). The second atmospheric CO_2 doubling from 2 × CO_2 to 4 × CO_2 leads to an additional warming of 7.4 K (Fig. 9). Hence, the climate sensitivity of the second CO_2 doubling is much larger than for the first CO_2 doubling. Heinemann et al. (2012), who investigated a range of CO_2 increases with the same model but in a Paleocene configuration, found that the larger climate sensitivity of the second doubling of atmospheric CO_2 is mainly caused by a larger

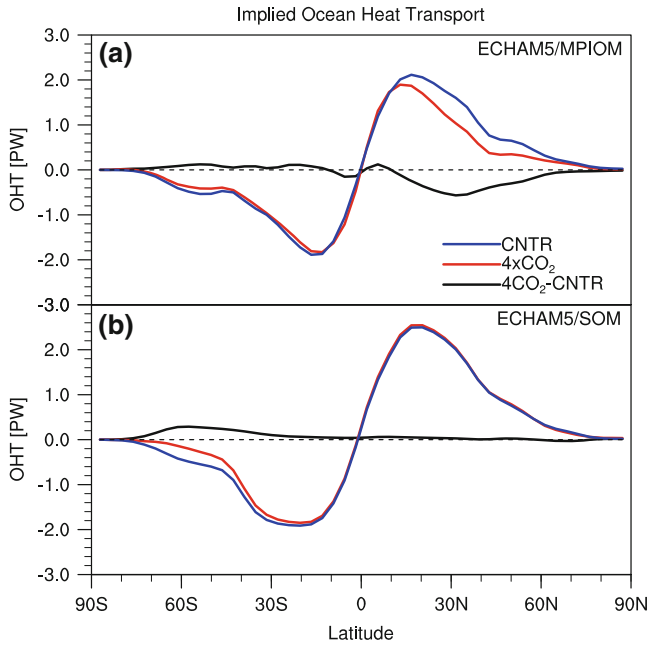


Fig. 8 Implied ocean heat transport simulated by **a** ECHAM5/MPIOM and **b** ECHAM5/SOM as a function of latitude for the $4 \times CO_2$ (*red lines*), control runs (*blue lines*) and the difference between $4 \times CO_2$ and control runs (*black line*). The heat transports are computed for the last 100 years of each run. Unit: 1.0 $PW = 1.0 \times 10^{15}$ W

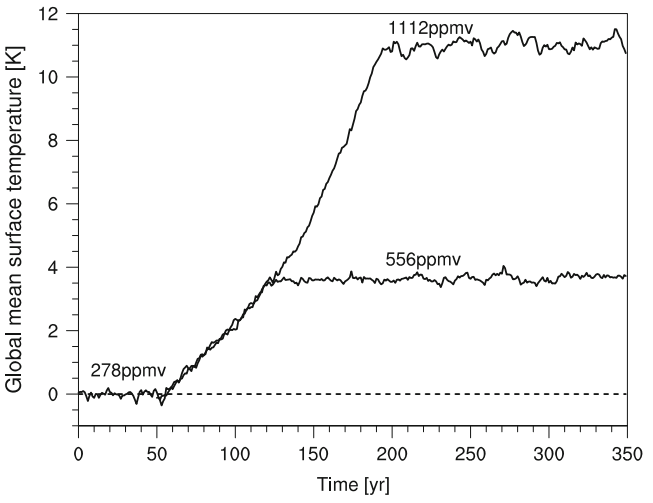


Fig. 9 Time series of global-mean surface temperature change of ECHAM5/SOM. All time series are relative to the pre-industrial control run of ECHAM5/SOM. Unit: K

Interesting

negative longwave cloud radiative forcing, leading to a warming of 4.6 K compared to 2.3 K for the first doubling. The larger surface albedo change and a larger reduction of the shortwave cloud radiative forcing together cause a warming of 1.9 K compared to 0.7 K for the first doubling.

While the global-mean surface temperature response to CO_2 quadrupling is very similar in ECHAM5/SOM and ECHAM5/MPIOM; the geographic distribution of the surface temperature change shows some marked differences between ECHAM5/SOM and ECHAM5/MPIOM (Fig. 10). ECHAM5/SOM shows a clearly asymmetric temperature change between Northern and Southern Hemispheres; the increase of the surface temperature in the Northern Hemisphere is much larger than that in the Southern Hemisphere (Fig. 10a). In the difference (ECHAM5/MPIOM minus ECHAM5/SOM), the maximum negative anomalies of over 5.0 K occur south of Greenland, corresponding to the typical deep convection site in the Labrador Sea. The negative anomalies extend eastward into most of the Eurasian continent. In the Southern Hemisphere, the surface warming in ECHAM5/MPIOM is larger than that in ECHAM5/SOM. The maximum positive anomalies of over 5.0 K occur over the Southern Ocean and the Antarctic continent. Generally speaking, this anomaly pattern between ECHAM5/MPIOM and ECHAM5/SOM is very similar to the bipolar seesaw response due to the AMOC reduction in the multi-model ensemble simulation of the water-hosing experiments (Fig. 14 in Stouffer et al. 2006), although there are differences in the amplitude in specific regions. The reduction in AMOC reduces the warming in the Northern Hemisphere but enhances the warming in the Southern Hemisphere, because of the associated reduction of the northward heat transport out of the Southern Hemisphere.

Shouldn't reduction in AMOC lead to greater surface warming?

However, different from the AMOC-induced bipolar seesaw pattern, we find in ECHAM5/MPIOM stronger warming of 3.0 K in the tropical Pacific and weaker warming in the western boundary regions in the North Pacific. In the tropical Pacific, the reduction of vertical heat transport, which is caused by the weakening of vertical mixing associated with the upper-ocean warming, enhances

Table 1 Globally averaged surface temperature in ECHAM5/MPIOM and ECHAM5/SOM, unit: K

	Global		Ocean	
	CNTR	$4 \times CO_2$	CNTR	$4 \times CO_2$
ECHAM5/MPIOM	286.7	297.5(+10.8)	288.9	298.9(+10.0)
ECHAm5/SOM	287.6	298.7(+11.1)	289.8	299.8(+10.0)

The numbers in parentheses in the last columns give the equilibrium surface temperature change in ECHAM5/MPIOM and in ECHAM5/SOM, relative to their pre-industrial control runs

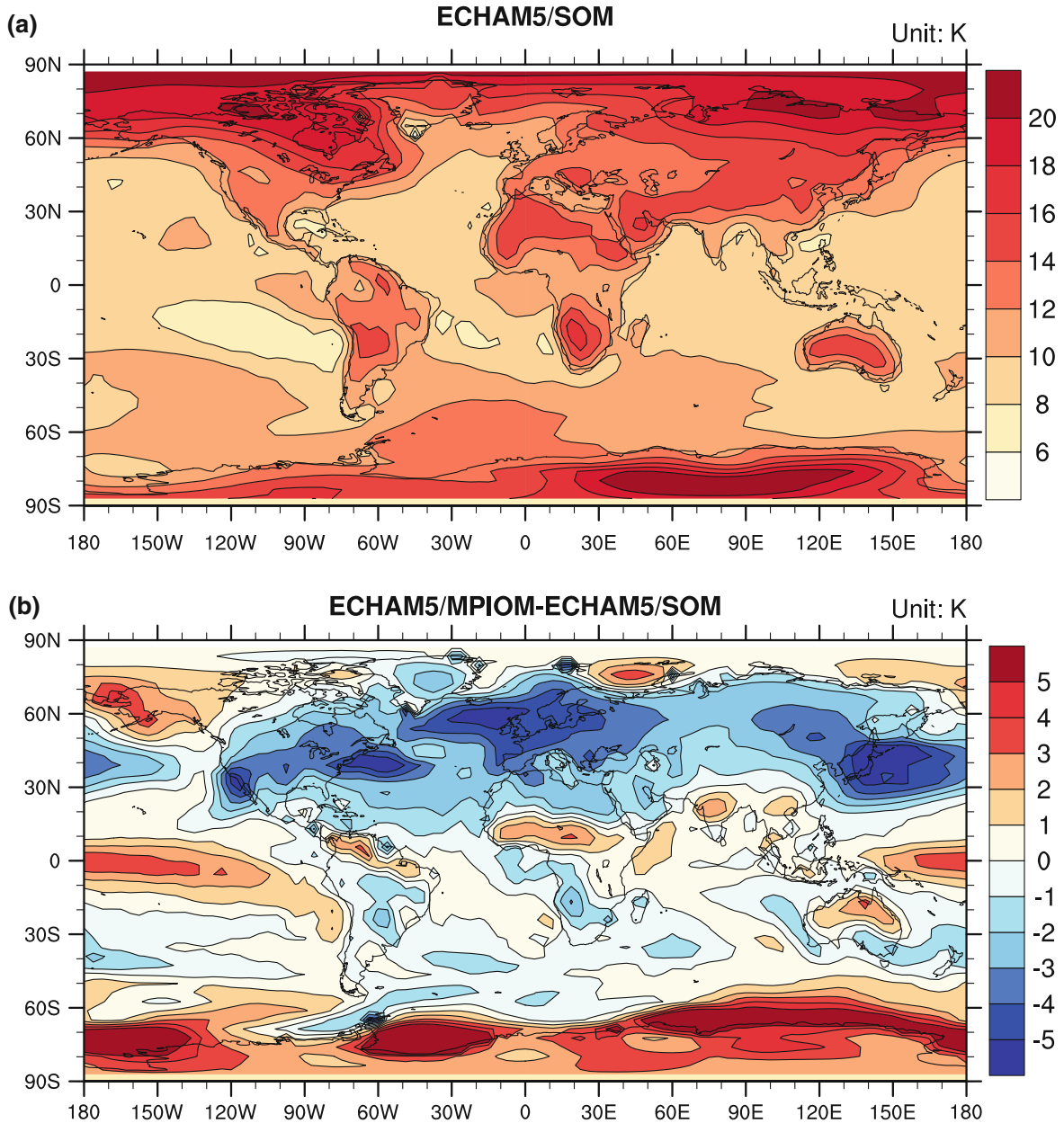


Fig. 10 Geographic distribution of the surface temperature change relative to the pre-industrial control run in response to CO_2 quadrupling in ECHAM5/SOM, and the difference of the geographic warming pattern between ECHAM5/MPIOM and ECHAM5/SOM. Unit: K

the surface warming found in the tropical Pacific. The tropical Pacific warming looks very similar to an El Nino-like response pattern and can influence the global climate through large-scale air-sea interaction. The tropical Pacific warming can cause the easterly trade winds to weaken and can thus weaken the Pacific western boundary current. Hence, the warming in western boundary current is reduced due to the reduction of the northward ocean heat transport. The ECHAM5/SOM cannot produce the enhanced tropical Pacific warming because the change in ocean circulation is not included in SOM. We also find much stronger warming over the Southern Hemisphere in the anomaly pattern between ECHAM5/MPIOM and ECHAM5/SOM, compared to the AMOC-induced bipolar seesaw pattern. The strong deep-ocean warming found in ECHAM5/MPIOM (details in Sect. 3b, c) can also enhance the surface warming because the Southern Ocean is very well mixed

5 Effective climate response

Extending the experiment to equilibrium enables us to test the method for estimating the effective climate response as described by Gregory et al. (2004). Here, we use the phrase

"effective climate response" instead of "effective climate sensitivity" to discriminate the long integration for CO_2 quadrupling from that for CO_2 doubling.

Following Gregory (2000) and Held et al. (2010), we use a two-layer horizontal-mean ocean model; the separation between the upper ocean and the deep ocean is at 1,500 m. We consider the following equations:

$$N = F - \alpha\Delta T \approx \Delta Q_S, \tag{4}$$

$$C_U\frac{dT_U}{dt} = \Delta Q_S - \Delta Q_D = F - \alpha\Delta T - \Delta Q_D, \tag{5}$$

$$C_D\frac{dT_D}{dt} = \Delta Q_D. \tag{6}$$

where N is the rate of increase in the heat stored in the climate system; because the ocean has a vastly larger heat capacity than the atmosphere, N approximately equals the total ocean heat uptake (ΔQ_S), and it also equals the net downward radiative flux at the top of the atmosphere (TOA). F is the radiative forcing, α is the climate response parameter, and ΔT is the global mean surface temperature change. ΔQ_D is the deep-ocean heat uptake (DOHU) defined as the tendency of total ocean heat content below 1,500 m. The unit of DOHU is W m^{-2} because we divide the total ocean heat content by the surface area of the Earth. ΔT_U and ΔT_D are the ocean mean temperatures in the upper and the deep ocean, respectively. C_U and C_D are the heat capacities of the ocean layers. We assume that the heat exchange is proportional to the difference between the temperature in the two layers,

$$\Delta Q_D = \lambda(\Delta T_U - \Delta T_D), \tag{7}$$

where λ is the deep-ocean heat uptake efficiency.

By using the data of the transient period, we perform a linear regression between N and ΔT, which gives us an estimate of both the radiative forcing F and the climate response parameter α. We find that the effective climate response ($\Delta Q_E = F/\alpha$, while $N = 0$) is 12.2 K, which is 1.4 K higher than the ECR of ECHAM5/MPIOM; the radiative forcing F is 5.26 W m^{-2}, and the climate response parameter α is 0.43 W m$^{-2}K^{-1}$. The fundamental assumption of this effective climate response method is the radiative forcing F and climate response parameter α are time-independent. However, Fig. 11 shows that the TOA net downward radiative flux decreases by 2.3 W m^{-2} with 4.9 K surface warming in the transient period (blue crosses), compared to a decrease by 0.7 W m^{-2} with 0.4 K surface warming in the quasi-equilibrium period (red crosses). The change in the slope of the data suggests that the climate response parameter α is time-dependent. However, while the steeper slope for the quasi-equilibrium period in Fig. 11 implies that the linear fit overestimates the equilibrium climate response; the error we commit is, at 1.4 K, which is relatively small.

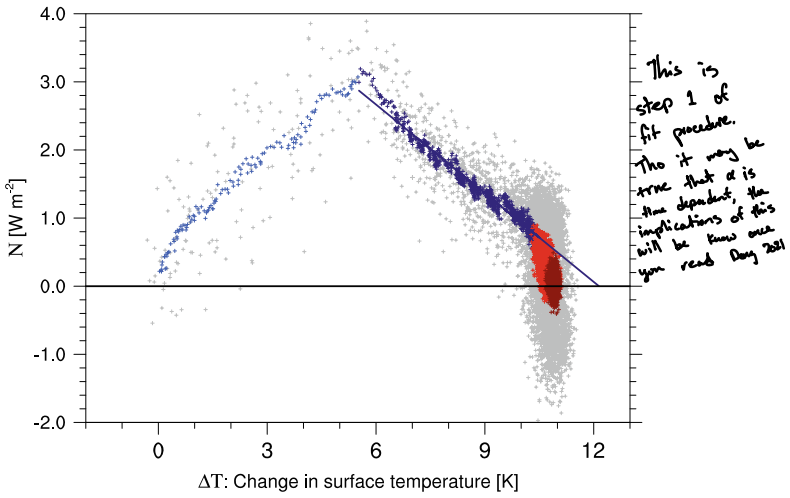


Fig. 11 Globally averaged TOA net downward radiative flux (N) (Unit: W m^{-2}) as a function of globally averaged surface temperature change (ΔT) (Unit: K). The *gray crosses* represent annual-mean data for the entire experiment, the other color crosses represent data from an 11-year running mean and relative to CNTR. The *light blue crosses* represent the CO_2 increase period (1–140 year), the *dark blue crosses* represent the transient period (140–1,200 year), the *red crosses* represent the quasi-equilibrium period (1,200–4,600 year), and the firebrick crosses represent the final equilibrium period (4,600–6,080 year). The *dark blue line* is a regression for the whole transient period

Our results suggest that the change in the climate response parameter α also plays an important role in the adjustment of the TOA radiative flux in the quasi-equilibrium period. According to (4), α can be diagnosed from $\alpha = (F - N)/\Delta T$. At the end of the transient period, there is still 0.7 W m^{-2} radiative flux imbalance at TOA; the climate response parameter change contributes 0.52 W m^{-2} ($\Delta\alpha \cdot \triangle\overline{T} = 0.05 \times 10.4 = 0.52\mathrm{W\,m^{-2}}$, where $\Delta\alpha$ is the climate response parameter change in the quasi-equilibrium period) to balance the radiative flux, whereas the surface temperature change (0.4 K warming in the quasi-equilibrium period) only contributes 0.17 W m^{-2} ($\alpha \cdot \triangle T' = 0.43 \times 0.4 = 0.17\mathrm{W\,m^{-2}}$, where $\triangle T'$ is the surface temperature change in the quasi-equilibrium period). This slow increase in α may be caused by the slow warming over the Southern Ocean and the Antarctic continent (Fig. 2a). By using the Hadley Centre coupled climate model with flux adjustment, Senior and Mitchell (2000) demonstrated that the time-dependence of the climate feedback parameter can be caused by cloud feedback arising from the inter-hemispheric temperature difference due to the slower warming of the Southern Ocean. Further investigations are still needed to quantify the impact of Southern-Ocean warming on the surface climate feedbacks in our experiment.

We can reduce the estimation error of the ECR still further by considering the upper and the deep ocean separately, noting that an alternative expression for the ECR

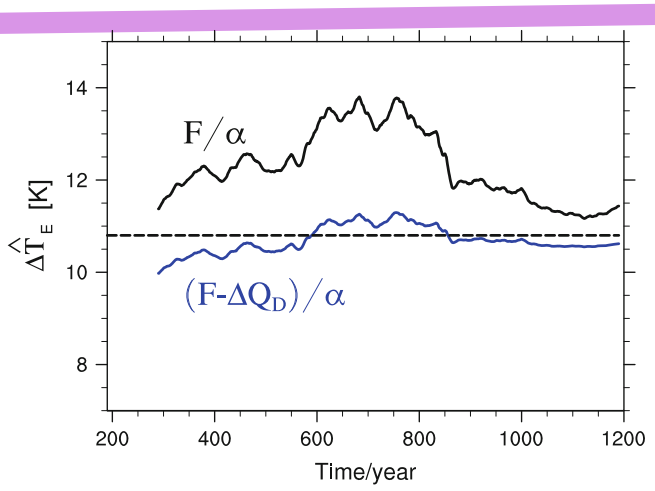


Fig. 12 Effective climate response. The *black solid line* represents the estimate following the method described by Gregory et al. (2004), the *blue solid line* represents the estimate including the deep-ocean heat uptake (ΔQ_D), and the *dashed black line* represents the ECR of ECHAM5/MPIOM, which is 10.8 K. A 300-year moving-window linear regression has been used. Unit: K

can be derived from (4) to (6), assuming $N = \Delta Q_D$ instead of $N = 0$,

$$\Delta \hat{T}'_E = (F - \Delta Q_D)/\alpha. \quad (8)$$

Interesting !!!

By using the data of the whole transient period (140–1,200 year), the ECR estimated by equation (8) is 10.7 K, which is very close to the ECR of 10.8 K in ECHAM5/MPIOM. To further clarify whether both methods work over the whole transient period, we use the same linear-regression technique with a sliding 300-year-window. The results suggest that the ECR estimated with the two-layer ocean model is much closer to the result from the ECHAM5/MPIOM simulation than the original method of Gregory et al. (2004) (Fig. 12). The overestimate by 10 % of the ECR method of Gregory et al. (2004) arises from ignoring the impact of DOHU on the surface climate response parameter.

6 Summary and conclusions

Using the coupled atmosphere-ocean-sea ice general circulation model ECHAM5/MPIOM, we perform a multi-millennium climate simulation by gradually increasing the CO_2 concentration by 1 % year^{-1} and keeping it constant after 140 years. The integration is continued until the whole system reaches equilibrium in year 6080. To our knowledge, we have achieved the first steady state simulation with CO_2 quadrupling by using a modern non-flux-adjusted AOGCM. The final equilibrium in the ECHAM5/MPIOM is compared with the corresponding simulation in ECHAM5/SOM (ECHAM5 coupled to a slab ocean model). We summarize our research as follows.

1. The equilibrium surface-temperature change from CO_2 quadrupling in ECHAM5/MPIOM is 10.8 K in the global mean and 10.0 K over the ocean.
2. The ocean temperature shows a near-uniform warming of around 8 K at almost all levels of the ocean; this result confirms globally averaged multi-box ocean model simulation (e.g. Harvey and Schneider 1985), but does not support the globally averaged upwelling-diffusion ocean model simulations (e.g. Harvey and Schneider 1985; Raper et al. 2001; Marčelja 2010).
3. We find that deep-ocean warming plays an important role for the thermosteric global sea level change. The globally averaged sea level still rises by 2.3 m due to the deep-ocean warming while the surface temperature is stationary. In the long term, surface temperature change is hence a poor predictor for steric sea-level change.
4. The equilibrium climate response in ECHAM5/SOM is 11.1 K, which is only 0.3 K higher than that in ECHAM5/MPIOM. This suggests that the change in AMOC and the reduced northward ocean heat transport have very limited effect on the global-mean surface temperature change, although the AMOC weakens in equilibrium by 46 %. However, the deep-ocean adjustment plays a very important role in determining the geographic pattern of equilibrium surface temperature response and its time evolution. The change in ocean heat transport damps the warming over the northern hemisphere mid and high latitudes, but enhances the warming over the tropical ocean and especially over the Southern-Hemisphere high latitudes. The pattern of the AMOC-induced surface temperature change in our experiments is similar to the multi-model ensemble simulation of water-hosing experiments (Stouffer et al. 2006), although there are differences in the amplitude in specific regions. In contrast to ECHAM5/SOM, which shows an asymmetric polar warming amplification, ECHAM5/MPIOM shows polar amplification in both the Arctic and Antarctic domains. The southern polar warming is greatly delayed by the Antarctic deep-ocean warming.
5. The equilibrium climate sensitivity in ECHAM5/MPIOM to CO_2 doubling amounts to 3.7 K, which is considerably less than half the ECR to CO_2 quadrupling (11.K). The climate sensitivity to the second CO_2 doubling (from 556 to 1,112 ppmv) is much larger than to the first CO_2 doubling (from 278 to 556 ppmv) because of a larger negative longwave cloud radiative forcing, a larger surface albedo change, and a larger reduction of the shortwave cloud radiative forcing (Heinemann et al. (2012).
6. The method to determine the effective climate response from transient simulation (Gregory et al.

2004) overestimates the ECR by only 10 %. The error is due to ignoring the impact of deep-ocean heat uptake on the estimated value of the surface climate feedback parameter.

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